课程大纲
COURSE SYLLABUS

1.	**课程代码/名称** **Course Code/Title**	**CSE5001 / 高级人工智能 Advanced Artificial Intelligence**
2.	**课程性质** **Compulsory/Elective**	**Compulsory**
3.	**课程学分/学时** **Course Credit/Hours**	**3/64**
4.	**授课语言** **Teaching Language**	**中文/ Chinese**
5.	**授课教师** **Instructor(s)**	**张宇/Yu Zhang**
6.	**是否面向本科生开放** **Open to undergraduates or not**	是/Yes
7.	**先修要求** **Pre-requisites**	（如面向本科生开放，请注明区分内容。 If the course is open to undergraduates, please indicate the difference.） **CS303 / 人工智能 Artificial Intelligence**

8. **教学目标**
Course Objectives

This course introduces recent advances in artificial intelligence. Topics covered include intelligent optimization and learning, as well as case studies in machine learning. The assessment in the course will consist of assignments, a mid-term test (or project), and a final exam.
Upon finishing this course, students are expected to have a good understanding of challenging optimization and learning problems in AI, and different models and algorithms for tackling these problems.

9. **教学方法**
Teaching Methods

Lectures and tutorials

10. **教学内容**
Course Contents

Section 1	Introduction
Section 2	Basic Search
Section 3	Heuristic Search
Section 4	Meta Heuristics
Section 5	Supervised Learning
Section 6	Ensemble Learning
Section 7	Multi-objective Learning
Section 8	Unsupervised Learning
Section 9	Feature Engineering
Section 10	Markov Decision Process

	Section 11	Reinforcement Learning
	Section 12	Natural Language Processing

11. 课程考核
Course Assessment

（①考核形式 Form of examination；②.分数构成 grading policy；③如面向本科生开放，请注明区分内容。If the course is open to undergraduates, please indicate the difference.）
Assignments: 40%
Mid-term test/Project: 20%
Final Exam: 40%

12. 教材及其它参考资料
Textbook and Supplementary Readings

Textbook: Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach (Third edition) , Cambridge University Press, 2009

Reading materials: Relevant papers as handed out at each lecture.